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PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Choroid Plexus and Hippocampal Damage and Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	ajwyrobek@lbl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	925-989-4466
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Rd MS: 74R0157		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2018: Key members are: Suzanne Celniker, Biosciences Area, Lawrence Berkeley National Laboratory (LBNL); Halina E. Witkowska (ret.), University of California, San Francisco; Bernard Rabin, University of Maryland, Baltimore Campus; Richard Britten, Eastern Virginia Medical School; Tore Straume, NASA Ames, Space Biosciences; Leif E. Peterson, Houston Methodist Research Institute; Oliver Fiehn, Matt Settles, Lutz and Froenicke, University of California, Davis.		
COI Name (Institution):	Peterson, Leif Ph.D. (The Methodist Hospital Research Institute) Straume, Tore Ph.D. (NASA Ames Research Center) Witkowska, Halina Ph.D. (University of California, San Francisco) Rabin, Bernard Ph.D. (University of Maryland, Baltimore Campus) Britten, Richard Ph.D. (Easterm Virginia Medical School) O'Banion, Kerry Ph.D. (University of Rochester) Celniker, Suzanne Ph.D. (University of California, Lawrence Berkeley National Laboratory)		
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Task Description:	A major goal of the NASA space radiation research program is to enable human exploration of space without exceeding limits for immediate and persistent risks to the central nervous system (CNS) from space radiation. Our research plan builds on the mounting evidence of the radiation-sensitivity of the hippocampus, which is important for memory and performance, and the choroid plexus (CP), which produces cerebral spinal fluid and maintains the blood-brain barrier. Using transcriptomic, proteomic, metabolomics, and immunohistochemical analyses of CNS tissue from rodents exposed to simulated space radiation, we have identified molecular and cellular changes associated with oxidative stress, neuro-inflammation, vasculature damage, changes in the protein composition of cerebral spinal fluid, and amyloid deposits such as those found in Alzheimer disease (AD). The ongoing research investigates the low-fluence responses and post-exposure-time dependence of these deficits, to identify the molecular mechanisms of inter-individual variations in behavioral responses after radiation exposure, and to validate the predicted mechanisms using tissue of irradiated and behaviorally characterized rodents. Our specific aims are to: (1) Investigate the cellular and functional abnormalities in the hippocampus and CP of HZE-irradiated rats that are predicted by our multi-omics findings with the emphasis on defects associated with AD. (2) Investigate hippocampal and CP radiation damage responses associated with inter-individual variation in radiation-induced neurocognitive deficits, using proteomics, RNA expression, metabolomics, and in situ CNS analyses, with emphasis on tissue defects associated with AD. (3) Characterize the radiation sensitivity of persistent CP epithelial and endothelial damage in AD mouse models and examine their associations with neuro-inflammation and other defects associated with AD. This research is foundational to establish quantitative mechanism-based predictors for neuro-cognitive risks in individuals exposed to high-LET (linear energy transfer) space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is designed to provide NASA with a mechanistic understanding of the risks for impaired neurocognitive performance (during mission) and individual susceptibility for persistent neuro-pathology and neurological diseases (after mission) after exposure to high energy heavy ions (HZE) radiation. Our approach: (a) employs state-of-the-art molecular profiling and tissue validation technologies to investigate CNS damage response in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with neurocognitive performance results, and (c) validates the predicted neuro-pathology in intact brain tissue. Our project focuses on radiation damage to the hippocampus (site of adult neuro-genesis; spatial memories; anxiety) and choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research strategy distinguishes between two broad categories of risk mechanisms: (1) mechanisms related to exposure dose, and (2) mechanisms related to inter-individual variations in radiation-induced CNS risks for neuro-cognitive performance (during mission) and risks for neurological diseases, i.e., Alzheimer Disease (AD). Our research is conducted in outbred strains of rats to capture the role of genetic diversity and in genetically defined double and triple mutation AD mice. Successful completion of this project will provide a mechanistic foundation for space-radiation-induced CNS risk that is based on quantitative molecular and cellular defects and their associations with measured neurocognitive and behavioral abnormalities. The project considers the time of onset and the persistence of effects, as well as inter-individual differences in responses. The project will employ space-relevant fluences of simulated HZE radiation, ensuring relevance for astronaut risk assessment. The results are expected to play a key role in helping to reduce the uncertainty associated with predicting the CNS risk for deep space exploration missions. This research is also designed to contribute to the identification of molecular targets for biological countermeasures to reduce CNS risks from space radiation.
	This project is designed to provide NASA with a mechanistic understanding of the risks from exposure to HZE radiation for impaired neuro-cognitive performance (during mission) and individual susceptibility for late-onset neurological diseases (after mission). Our approach: (a) employs state-of-the-art molecular profiling and tissue validation technologies to investigate CNS damage response in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) links the type and magnitude of molecular defects with neuro-cognitive performance, and (c) validates the predicted molecular neuro-pathology in brain tissue. Our project focuses on radiation damage to the hippocampus (site of spatial memory and anxiety responses) and choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research strategy distinguishes between two broad categories of risk mechanisms: (1) mechanisms related to exposure dose, and (2) mechanisms of inter-individual variations in risks for neuro-cognitive performance during the mission and late-onset neurological diseases (i.e., Alzheimer's Disease). Male rats were exposed to graded doses of 1GeV/n 56Fe at NASA Space Radiation Laboratory (NSRL) (10 - 100 cGy), and tested for novel object memory, spatial memory, operant behavior, movement, and anxiety. Tissues for molecular profiling (proteomic, transcriptomic, and metabolomics) from cerebral spinal fluid (CSF), choroid plexus (CP), cortex, and hippocampus were collected between 3 days to 22 months after exposure to generate hypotheses of predicted neuro-pathologies. These predictions for abnormal behaviors and persistent neuro-pathologies were validated by direct comparisons between each animal's behavior performance, molecular profile, and histopathology. Tissue validation of neuro-pathologies predicted by molecular profiling: Analyses of changes in molecular profiles across animals' lifetimes in CSF, hippocampus and cortex tissues of irradiated animals predicted several neuro-pathologies of which three were validated in tissue sections (molecular histopathology, endothelial fibrosis, and amyloid deposition). Exposures as low as 10cGy 56Fe produced expression abnormalities in CSF proteins and transcripts at 2 and 22 months (m) after exposure, respectively. Affected genes were associated with neurological diseases, vascular defects, and amyloid/fibril abnormalities, and selected predictions were tested in tissue sections of the same animals. Beginning at ~ 4m and increasing at 9 and 22 m after exposure, high-LET irradiated brains showed: (a) abnormalities in CP epithelial histology and reduced epithelial expression of transthyretin (TTR), a CSF protein associated with ß-amyloid clearance, (b) increased CP endothelial vascular fibrosis at 9, 15, and 22 m after exposure, and (c) increased frequencies of amyloids (Congo Red), extracellular aggregations of amyloid beta A4, and increased extracellular TTR and ß-amyloid bodies throughout the CNS at 9, 15 and 22 m after exposure. Our findings suggest that CP and CSF are sensitive targets of radiation-induced changes in the expression of genes associated with abnormal behavior and increased risks for late-onset CNS neuro-pathologies.

Task Progress:	Characterization of individual and genotype variations in behavioral performance of animals exposed to high LET beams: Spatial memory learning in Wistar rats: We investigated the dose responses for spatial memory learning among genetically outbred male Wistar rats exposed to graded doses of 56Fe particles (sham, 5, 10, 15, and 20 cGy; 1 GeV/n). Spatial memory learning was assessed at 3 months after exposure on a Barnes maze. In Wistar rats, we observed a monotonic decline in learning across these doses along a linear response. Inspection of the variation in performance at each dose suggested that individuals vary substantially in spatial memory learning and that exposures in this low-dose range appear to predominantly impact poor performers. The proportions of poor learners and the magnitude of their impairment followed linear responses with doubling doses of ~10 cGy. In contrast, good learners showed no detectable learning deficits in this dose range. These observation invites future investigations of the underlying physiologic and genetic mechanisms of inter-individual susceptibility in spatial memory performance after low-dose HZE radiation exposures. Comparison of three behavioral endpoints among Sprague Dawley rats exposed to 56Fe particle beams: Male outbred Sprague Dawley rats were irradiated at NSRL (56Fe, 1GeV/n; sham, 10-20cGy; n=40-44; 2-m old at exposure) and evaluated for three behavior endpoints by video software on open field platform, elevated plus maze and Barnes maze. Exposures of 10 and 20 cGy 56Fe induced changes in spatial memory and anxiety-related endpoints at 1-3 months after exposure with no detectable effects on animal movement. The observation of differences in responses across behavior endpoints suggests that CNS behavior networks may differ in their radiation sensitivity. Genotype comparison of spatial memory performance in Sprague Dawley versus Wistar rats exposed to 56Fe particle beams: The comparison of Barnes-maze performance of ~160 male Wistar and ~130 male Sprague Dawley rats after exposure in the dose range up to 20cGy identified significant differences in the radiation responses in the behaviors of the two genotypes: time of shams to find the escape hole across the six trials of the assay, whether radiation accelerated or impeded the time to find the escape hole after exposure, and the differences in learning across the six trials of the Barnes maze protocol. This study emphasizes the importance of considering genetic background when assessing radiation effects on behavior, and further controlled studies are needed using both behavior and molecular endpoints. Molecular pathways associated with extremes in anxiety before and after exposure to 56Fe particle beams: We compared proteomic profiles of choroid plexus and hippocampus of anxiety-characterized animals exposed to 56Fe particles. Male Sprague Dawley rats, exposed at 6 months of age to two doses of 56Fe particles (1 GeV/n; sham, 10cGy and 100cGy; n=24 per group), were evaluated ~1 m after exposure on an elevated plus maze. Radiation exposure significantly increased closed-arm preference in each dose group. The proteomic profiles of CP and hippocampal tissues collected from animals at the extremes of the anxiety distributions in each dose group showed significant correlations with baseline anxiety, independent of exposure, and these profiles were significantly associated with upstream regulation by MAPT, PSEN1, APP, and HTT, which are known risk factors for Alzheimer disease and other neurological abnormalities. In contrast, proteins showing significant correlation with anxiety only in irradiated animals were associated with neuro-inflammation, neurotangle and amyloid biology, certain neurotransmitters, and synaptic activity. Confirmatory experiments are in progress. Working model of the tissue mechanisms that predict individual variation in behavioral deficits and risks for late-onset neurological abnormalities in irradiated animals: We are building a mechanistic model of molecular predictions for abnormal behavior and increased risks for neurological disease after exposure to high-LET radiation. We have reported that the CP and CSF, which are major components of the blood-CSF brain barrier, are sensitive to exposure to high-LET exposures. Certain radiation-induced molecular abnormalities persist long after exposure, in some cases for the life of the animal. Persistent abnormalities in expression predict increased risks for neuro-pathologies, such as abnormal CSF composition, vascular abnormalities, and abnormal formation of protein aggregates and amyloids. Our model incorporates recent findings that inter-individual variations in behavior outcomes are correlated with molecular profile changes in CNS subregions, offering a promising approach to identify individuals that may be more susceptible to low-dose HZE exposures. In our model, individual susceptibility to radiation induced anxiety responses and memory deficits are associated with the differential expression of CP and hippocampal proteins, which provides a novel window into understanding molecular mechanisms that may predict individual variations in CNS responses to space radiation exposures and may lead to the future development of mechanism-based molecular countermeasures.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Wyrobek AJ. "Individual variations in behavior and neuro-pathology after early-life exposure to space radiation." Invited presentation. Oregon Health Sciences University, June 29, 2017. Invited presentation. Oregon Health Sciences University, June 29, 2017. , Jun-2017
Abstracts for Journals and Proceedings	Wyrobek AJ. "Radiation-reduced expression of transthyretin (TTR) in the choroid plexus is associated with increased CNS vascular fibrosis, amyloid formation, and anxiety behaviors." Invited presentation. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Wyrobek AJ. "Persistent behavioral deficits and molecular neuro-pathologies after exposures to heavy ion radiation." Invited lecture series presentation. Kavli Faculty Seminar, Genentech Hall, Univ of California, San Francisco, January 8, 2018. Kavli Faculty Seminar, Genentech Hall, Univ of California, San Francisco, January 8, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Wyrobek AJ. "Molecular Characterization of Choroid Plexus and Hippocampal Damage and Degenerative CNS Risks from Exposure to Space Radiation on Flights to Mars." Invited presentation. 2017 UC Berkeley Neuroscience Annual Conference, Granlibakken Conference Center, California, October 13-15, 2017. 2017 UC Berkeley Neuroscience Annual Conference, Granlibakken Conference Center, California, October 13-15, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Wyrobek AJ. "Anxiety-associated protein expression in the choroid plexus and hippocampus in an outbred rat model." Invited presentation. Cellular and Endocrine Basis for the Effects of Stress on the Brain. Neuroscience 2017, Washington, DC, November 11-15, 2017. Neuroscience 2017, Washington, DC, November 11-15, 2017. , Nov-2017

Abstracts for Journals and Proceedings	Wyrobek A, Rabin B, Britten R, Bhatnagar S, Peterson L, Straume T, Witkowska H. "Molecular characterization of choroid plexus and hippocampal damage and degenerative CNS risks from space radiation." Invited presentation. 29th Annual Space Radiation Investigators' Workshop and 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 29th Annual Space Radiation Investigators' Workshop and 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018